SANGAY WANGCHUK (PhD)

ASSISTANT PROFESSOR (CHEMISTRY)

Sherubtse College, Royal University of Bhutan, Kanglung, Bhutan

+97517658561

swangchuk.sherubtse@rub.edu.bt

ABOUT ME

PhD, Chemist & Educator with specialization in electrochemical sensor research, teaching, and curriculum design. Strong experience in mentoring students, solving complex problems, and publishing research. Skilled at adapting teaching methods to

RESEARCH INTEREST

- Electroanalytical Chemistry
- Chemical Sensors & Biosensors
- Electrochemical analysis
- Forensic electrochemistry
- Drug & biomedical analysis
- Nanomaterials & Nano catalysts synthesis

LINKS

ORCID:
https://orcid.org/0000-0001-7606-3399

RESEARCHGATE:
https://www.researchgate.net/profile/Sangay-Wangchuk-13

SCOPUS:
https://www.scopus.com/authid/detail.uri?authorId=58100180900

GoogleScholar:
https://scholar.google.com/citations?user=wZITV2EAAAAJ&hl=en&oi=ao

WORK EXPERIENCE

SHERUBTSE COLLEGE, ROYAL UNIVERSITY OF BHUTAN
Kanglung
Feb 2019 - Present

Assistant Professor (Chemistry)
- Developed undergraduate chemistry curriculum
- Led hands-on lab sessions
- Guided student research & careers
- Assessed & updated course contents
- Developed undergraduate Chemistry modules

MINISTRY OF EDUCATION
Jan 2008 – Jan 2019

Teacher (Chemistry)
- Taught High School chemistry
- Designed lessons & labs
- Personalized student aid
- Contributed to board examination question setting and test development
- Developed exams & curriculum

EDUCATION

PRINCE OF SONGKLA UNIVERSITY
Hat Yai
2024

PhD
- PhD with excellent academic standing (94%)
- Pioneered sensor design for diverse applications
- Boosted sensor acuity with novel carbon nanomaterials & metal nanoparticles
- Mastered electrode tuning & elevating performance metrics
- Published impactful research in peer-reviewed international journals

ANDHRA UNIVERSITY
Visakhapatnam
2012

Master of Science
- Specialized in inorganic synthesis
- Adept in metal complex synthesis & analysis
- Advanced in coordination chemistry
- Achieved MSc Chemistry with distinction

SAMTSE COLLEGE OF EDUCATION
Samtse
2007

Post Graduate Certificate in Education
- Honed advanced teaching & classroom management skills
- Skilled in evaluation & assessment tactics
- Expertise in diverse instructional techniques
- Real-classroom application of pedagogy

SHERUBTSE COLLEGE
Kanglung
2005

Bachelor of Science
- Acquired a core understanding of Chemistry, Botany, and Zoology
- Honed lab competencies
- Fulfilled coursework in the above fields
- Executed experiments, bridging theory with hands-on applications

SKILLS

- RESEARCH DESIGN
- RESEARCH AND ANALYSIS
- ELECTROCHEMICAL DATA ANALYSIS
- RESEARCH COLLABORATION
- APPLICATION OF ELECTROCHEMICAL TECHNIQUES
- TEACHING

PROFESSIONAL SERVICES

Program Leader, BSc in Chemistry, Sherubtse College, Royal University of Bhutan (March 2025 – present)

Principal Investigator: PI of a research project on Water Source Sampling and Water Quality Testing for Dorjilung Hydropower Project Limited, funded by the Druk Green Power Corporation (DGPC, 2026).

Principal Investigator: PI of a research project on Environmental Assessments for the 400 kV D/C East–West Transmission Line project, funded by the Bhutan Power Corporation (BPC, 2025).

Principal Investigator: PI of a research project on Environmental Assessments for the 132 kV D/C Gamri I & II Transmission Line projects, funded by the Bhutan Power Corporation (BPC, 2025).

Co-principal Investigator: Co-PI of a research project on Real-Time Monitoring of Drinking Water Quality at Sherubtse College Using IoT Sensors and Laboratory Validation (STLRG, 2025-2026).

Reviewer, Chemistry Textbook XII, Cambridge Curriculum, Ministry of Education and Skills Development (MoESD), Jampelyang Hotel, Haa (July 6-12, 2026)

Reviewer, Chemistry Textbook XI, Cambridge Curriculum, Ministry of Education and Skills Development (MoESD), Sherabgatshel Primary School, Thimphu (December 9-13, 2025)

Reviewer, Bhutan Journal of Research & Development (ISSN:2072-9065, 2025)

Editorial Member: Editorial member of the national journal 'Sherub Doenme', Sherubtse College, Royal University of Bhutan (August 2025- present)

Curriculum Reform Facilitator: Delivered a keynote presentation on Program Quality Control (PQC) recommendations to 25 STEM faculty members, guiding the integration of feedback into the revised STEM curriculum during a strategic retreat at Lingkhar Resort (June 6, 2025).

Program Review: Spearheaded a 6-day faculty workshop at Lingkhar Resort (June 6-11, 2025) to redesign the BSc in Chemistry program, ensuring alignment with international academic and industrial standards.

Reviewer, Bhutan's Class IX Chemistry FlexBook®, CK-12 Foundation (USA), HM Secretariat, and Ministry of Education and Skills Development (MoESD), under the "Empowering Bhutan's Education" initiative, 2025. Bhutan Team.

PhD Talk: Shared PhD Journeys and Research Experiences-Organized by the Office of DRIL and DAA, Sherubtse College, Royal University of Bhutan, this event brought together faculty and undergraduate students from STEM to explore academic pathways, address research challenges, and foster collaborative opportunities (April 9, 2025)

Reviewer, National Journal Sherub Doenme (ISSN:1027-0922, 2025)

Reviewer, International Journal of Nature, Environment and Pollution Technology (ISSN: 2395-3454)

Program review: Reviewed and developed four-year undergraduate modules for the BSc in Chemistry programme, Sherubtse College, Royal University of Bhutan at Karmaling, Trashi Yangtse (20-23 February 2025)

Reviewer: Environmental Toxicity, Human Hazards, and Bacterial Degradation of Polyethylene. Nature Environment and Pollution Technology, 2023

Program review: Reviewed and developed four-year undergraduate modules for the BSc in Chemistry programme, Sherubtse College, Royal University of Bhutan at Druk Doethjung, Trashigang (5-11 July 2021)

Program Leader, BSc in Chemistry, Sherubtse College (2021)

Editorial Board Member, Sherubtse College Magazine, Sherubtse College (March 2020 – 2021)

Co-PI of the research project: Detection of Adulterants in Common Food Items Available in the Bhutanese Market, funded by the Annual College Research Grant, Sherubtse College, Royal University of Bhutan (2020-2021)

Core Member of the research project: Water Quality and Aquatic Ecology Assessment for the 20 MW Yungichhu Hydropower Project, funded by DGPC (2021)

Core Member of the research project: Water Quality and Aquatic Ecology Assessment for the 8 MW Thungdiri Hydropower Project, funded by DGPC (2021)

Core Member of the research project: South Asian Nitrogen Hub (SANH)-WP3.1: Role of Nitrogen Air Pollution on Forest Ecosystem Services, funded by UKRI-GCRF (2020-2021)

Member of the Assessment Committee for Research Proposal Defense, BSc in Chemistry with Honours, Sherubtse College (2021)

Core Member of the research project: Hydrograph Separation of Streamflow Using Geochemical Tracers in Paa Chu Basin, funded by the World Bank (2019-2020)

Developed Laboratory Manuals for the BSc in Chemistry programme, Sherubtse College, Royal University of Bhutan (2019-2020)

Academic Secretary, Rangjung Higher Secondary School, Trashigang (2017 – January 2019)

Chief Marker, BHSEC Chemistry (2018), College of Science and Technology, Phuntsholing, organized by the BCSEA

RESEARCH GRANTS

Secured Ngultrum 522,588.00 from Dorjilung Hydropower Project Limited (DHPL) for the consultancy project titled Water Source Sampling and Water Quality Testing.

Secured Ngultrum 3,714,150.00 from Bhutan Power Corporation (BPC) for the consultancy projects titled Environmental Assessments for the 400 kV D/C East–West Transmission Line project, funded by the Bhutan Power Corporation (BPC, 2025).

Secured Ngultrum 1,027,950.00 from Bhutan Power Corporation (BPC) for the consultancy projects titled Environmental Assessments for the 132 kV D/C Gamri I & II Transmission Line projects.

Secured Ngultrum 498239.00 Sherubtse Thorim Lobdra Research Grant for the research project on Real-Time Monitoring of Drinking Water Quality at Sherubtse College Using IoT Sensors and Laboratory Validation (STLRG, 2025-2026).

Secured Ngultrum 200,000 from the Annual College Research Grant (ACRG Mid-level Grant 2020) for the project: Detection of Adulterants in Common Food Items Available in the Bhutanese Market.

AWARDS AND SCHOLARSHIPS/FINANCIAL SUPPORT

Received **Outstanding Thesis Award 2025** for the thesis titled "Development of Electrochemical Sensing Platforms Based on Functionalized Carbon Nanomaterials: Applications in Environmental Monitoring and Biomedical Analysis" from Prince of Songkla University, Thailand

2021-2024 Center of Excellence for Trace Analysis and Biosensor (TAB-CoE) **Scholarship** to pursue a PhD in Electrochemical Sensors, Prince of Songkla University, Thailand

2021-2024 Prince of Songkla University, Graduate Studies Financial **Scholarship** for Thesis, Prince of Songkla University (PSU), Thailand

2021-2024 The Royal University of Bhutan (RUB) **Scholarship** to pursue a PhD in Thailand

Financial Support from the National Science, Research and Innovation Fund (NSRF) and Prince of Songkla University, Thailand (Grant No SCI6701386a) for the project titled: Electrode modified with CO_2 laser-reduced graphene oxide-silver nanoparticles for determination of nitrite in water.

Financial Support from the National Science, Research and Innovation Fund (NSRF) and Prince of Songkla University, Thailand (Grant No SCI6701386a) for the project titled: A portable disposable metal-free electrochemical sensor for uric acid measurement in human blood serum.

Financial Support from the National Science, Research and Innovation Fund (NSRF) and Prince of Songkla University, Thailand (Grant No SCI6701386d) for the project titled: A disposable metal-free electrochemical sensor uses a boron/nitrogen co-doped multi-walled carbon nanotubes electrocatalyst to determine the anticancer drug flutamide.

Financial Support from the National Science, Research and Innovation Fund (NSRF) and Prince of Songkla University, Thailand (Grant No SCI6701386c) for the project titled: Cuprous oxide-functionalized activated porous carbon-modified screen-printed carbon electrode integrated with a smartphone for portable electrochemical nitrate detection.

Royal Civil Service Award (2018) for completing ten years of teaching service in Bhutan.

SKILLS AND COMPETENCIES

TECHNICAL SKILLS

[1]. CO_2-laser machine operations
[2]. Proficient in PalmSens Software operations
[3]. Statistical analysis (TWO-WAY ANOVA)
[4]. Expert in experimental data interpretation related to SEM, EDS, FTIR, XRD, Raman spectroscopy, and UV-Vis spectrophotometry
[5]. Expertise in the analysis of physicochemical parameters, nutrients, and trace metals in river systems
[6]. Proficient in statistical evaluation of water quality data to identify trends, assess regulatory compliance, and support evidence-based environmental decision-making
[7]. Apply a scientific approach to problem-solving, effectively communicate findings through models, charts, and graphs, and present research outcomes using clear and precise scientific writing

LABORATORY SKILLS

[1]. Synthesis of hybrid nanomaterials for electrode fabrication using hydrothermal, electrodeposition, chemical activation, green synthesis, microwave-assisted synthesis, and calcination techniques
[2]. Synthesis of composite nanomaterials using CO_2-laser irradiation
[3]. Fabrication of electrodes using the CO_2-laser patterning technique
[4]. Modification of the electrode surface using different hybrid nanomaterials

[5]. Extensive knowledge in the operation of the Potentiostat and UV-Vis spectrophotometer
[6]. Preparation of reagents and standards, sample preparation, calibration, operation, and data processing using analytical instruments such as Palmsens potentiostat, DO meter, pH meter, electrical conductivity meter, turbidity meter, COD, and UV-Vis spectrophotometer

COMPUTER SKILLS

[1]. Proficient in MS Office, Microsoft Excel, and PowerPoint
[2]. Additional skills include Origin 18 Graphing and Analysis, and ChemDraw Professional 19

PERSONAL SKILLS

[1]. Strong interpersonal skills with the ability to effectively collaborate in a team environment and work independently
[2]. Adaptable and proactive, able to manage multiple tasks efficiently while maintaining attention to detail
[3]. Problem-solving mindset, capable of thinking critically and approaching challenges with a solutions-oriented approach
[4]. Strong work ethic, committed to meeting deadlines and achieving high standards of quality in all tasks
[5]. Proven ability to design, conduct, and lead scientific research with applications in biomedical, environmental monitoring, forensic, and food safety

LANGUAGE AND COMMUNICATION SKILLS

[1]. Proficient in English, both spoken and written
[2]. Documented ability to use English effectively for writing scientific publications and research grant proposals, and for fluent communication.
[3]. Fluent in Dzongkha, with excellent spoken and written communication skills

RESEARCH PUBLICATIONS

Peer-Reviewed Journal Articles

[1]. Promsuwan, K., S., Saichanapan, J., Soleh, A., Saisahas, K., Samoson, K., **Wangchuk, S.**, & Limbut, W. (2026). A smartphone-assisted NFC-enabled microfluidic electrochemical sensor for on-site monitoring of residual free chlorine in water. *Journal of Environmental Chemical Engineering*. (**Q1, IF 7.5**). https://doi.org/10.1016/j.jece.2026.123838.

[2]. Promsuwan, K., Sanguarnsak, C., Kongsuwan, A., Chuenjitt, S., Saichanapan, J., Soleh, A., Saisahas, K., Samoson, K., **Wangchuk, S.**, & Limbut, W. (2026). Ammonium Ion-Sensing Skin Patch Based on Three-Dimensional Nanoarchitecture of Polystyrenesulfonate: Polyaniline/Copper Microflowers Deposited on Flexible Graphene Electrodes. *ACS Applied Nano Materials*. (**Q2, IF 5.5**). https://doi.org/10.1021/acsanm.5c05854.

[3]. Saisahas, K., Soleh, A., Samoson, K., Dechthongchai, M., Promsuwan, K., Saichanapan, J., **Wangchuk, S.**, & Limbut, W. (2026). Electrochemical clonazepam sensor based on B-doped laser-induced graphene for on-site forensic analysis. *Microchemical Journal*, 117476. (**Q1, IF 5.1**). https://doi.org/10.1016/j.microc.2026.117476

[4]. Soleh, A., Saisahas, K., Samoson, K., Phetrat, D., Promsuwan, K., Saichanapan, J., **Wangchuk, S**., Somapa, N., Somapa, D., & Limbut, W. (2025). N-doped porous laser-induced graphene applied for forensic electrochemical sensing of xylazine. *Microchemical Journal*, 114935. (**Q1, IF 5.1**). https://doi.org/10.1016/j.microc.2025.114935

[5]. Samoson, K., Saisahas, K., Soleh, A., Martkiattikul, S., Promsuwan, K., Saichanapan, J., **Wangchuk, S**., ... & Limbut, W. (2025). One-step laser fabrication of a P-doped 3D porous graphene electrode for on-site detection of promethazine. *Talanta*, 128715. (**Q1, IF 6.1**). https://doi.org/10.1016/j.talanta.2025.128715

[6]. Soleh, A., Saisahas, K., Samoson, K., Promsuwan, K., Saichanapan, J., **Wangchuk, S**.,
Somapa, N., Somapa, D., & Limbut, W. (2025). Revolutionizing oil palm biomass into laser-induced graphene for sustainable and high-performance electrochemical sensors. *Materials Today Chemistry*, *47*, 102872. (**Q1, IF 6.7**). https://doi.org/10.1016/j.mtchem.2025.102872

[7]. Saisahas, K., Soleh, A., Samoson, K., Promsuwan, K., Saichanapan, J., **Wangchuk, S.**, & Limbut, W. (2025). Sustainable Paper-Derived Laser-Induced Graphene Electrochemical Platform for Ultra-Sensitive Diazepam Detection in Forensic Investigations. *ACS Omega, 10*(28, 30944–30957. (**Q1, IF 4.3**). https://doi.org/10.1021/acsomega.5c03662

[8]. Promsuwan, K., Sanguarnsak, C., Kongsuwan, A., Saichanapan, J., Soleh, A., Saisahas, K., Samoson, **Wangchuk, S.,** ... & Limbut, W. (2025). Smartphone-enabled detection of urea in animal feed based on a disposable electrode modified with silver nanoparticles decorated on nitrogen-doped graphene nanoplatelets. *Talanta*, 128431. (**Q1, IF 6.1**). https://doi.org/10.1016/j.talanta.2025.128431.

[9]. Promsuwan, K., Chuenjitt, S., Kongsuwan, A., Saichanapan, J., Soleh, A., Saisahas, K., Samoson, K., **Wangchuk, S.**, & Limbut, W. (2025). A disposable dual-mode electrochemical/colorimetric paper-based analytical device for simultaneous detection of hydroquinone and mercury ion. *Talanta, 294*, 128166. (**Q1, IF 6.1**). https://doi.org/10.1016/j.talanta.2025.128166.

[10]. Somsiri, S., Kaewjangwad, C., Malarat, N., **Wangchuk, S.**, Saichanapan, J., Samoson, K., ... & Limbut, W. (2025). Portable NFC potentiostat integrated with a

3D paper-based microfluidic electrochemical device for glucose detection in whole blood using PEDOT:PSS/DMSO/GOx sensitive film. *Microchemical Journal*, *213*, 113623. (**Q1, IF 5.1**). https://doi.org/10.1016/j.microc.2025.113623.

[11]. Sanguarnsak, C., Promsuwan, K., Samoson, K., Saichanapan, J., Soleh, A., Saisahas, K., **Wangchuk, S.,** .... & Limbut, W. (2025). A β-cyclodextrin/porous graphene ink electrode for smartphone-assisted electrochemical Hg^{2+} sensing. *Talanta, 292*, 127776. (**Q1, IF 6.1**). https://doi.org/10.1016/j.talanta.2025.127776.

[12]. Samoson, K., Saisahas, K., Soleh, A., Promsuwan, K., Saichanapan, J., **Wangchuk, S.**, ... & Limbut, W. (2025). NS Dual-Doped 3D Porous Laser-Induced Graphene Electrode for Curcumin Determination in Turmeric. *Talanta*, *288*, 127722. (**Q1, IF 6.1**). https://doi.org/10.1016/j.talanta.2025.127722.

[13]. **Wangchuk, S.**, Promsuwan, K., Saichanapan, J., Soleh, A., Saisahas, K., Samoson, K., ... & Limbut, W. (2025). Cuprous oxide-functionalized activated porous carbon-modified screen-printed carbon electrode integrated with a smartphone for portable electrochemical nitrate detection. *Talanta*, *287*, 127581. (**Q1, IF 6.1**). https://doi.org/10.1016/j.talanta.2025.127581.

[14]. Promsuwan, K., Saichanapan, J., Soleh, A., Saisahas, K., Samoson, K., **Wangchuk, S.,** & Limbut, W. (2025). Portable unibody semi-flow injection voltammetric sensor for on-site screening of illegal additive sibutramine in food supplements. *Talanta*, *283*, 127123. (**Q1, IF 6.1**). https://doi.org/10.1016/j.talanta.2024.127123.

[15]. **Wangchuk, S.,** Promsuwan, K., Saichanapan, J., Soleh, A., Saisahas, K., Samoson, K., ... & Limbut, W. (2024). A disposable metal-free electrochemical sensor uses a boron/nitrogen co-doped multi-walled carbon nanotubes electrocatalyst to determine the anticancer drug flutamide. *Microchemical Journal*, *207*, 112217. (**Q1, IF 5.1**). https://doi.org/10.1016/j.microc.2024.112217.

[16]. **Wangchuk, S.**, Promsuwan, K., Saichanapan, J., Soleh, A., Saisahas, K., Samoson, K., ... & Limbut, W. (2024). A portable disposable metal-free electrochemical sensor for uric acid measurement in human blood serum. *Microchemical Journal*, *207*, 112216. (**Q1, IF 5.1**). https://doi.org/10.1016/j.microc.2024.112216.

[17]. Promsuwan, K., Saichanapan, J., Soleh, A., Saisahas, K., Samoson, K., **Wangchuk, S.,** & Limbut, W. (2024). Bio-functionalized conductive poly (acrylic acid): poly (3, 4-Ethylenedioxythiophene)-Prussian blue hybrid transducer for biosensors and bioelectronics interfaces. *Materials Today Chemistry*, *40*, 102271. (**Q1, IF 6.7**). https://doi.org/10.1016/j.mtchem.2024.102271.

[18]. **Wangchuk, S.,** Promsuwan, K., Saichanapan, J., Soleh, A., Saisahas, K., Samoson, K., ... & Limbut, W. (2024). Electrode modified with CO2 laser-reduced graphene oxide-silver nanoparticles for determination of nitrite in water. *Microchemical Journal*, *204*, 111100. (**Q1, IF 5.1**). https://doi.org/10.1016/j.microc.2024.111100.

[19]. Promsuwan, K., Sanguarnsak, C., Samoson, K., Saichanapan, J., Soleh, A., Saisahas, K., **Wangchuk, S.,**... & Limbut, W. (2024). Single-drop electrodeposition of nanoneedle-like bismuth on disposable graphene electrode for on-site electrochemical detection of cadmium and lead. *Talanta*, *276*, 126179. (**Q1, IF 6.1**). https://doi.org/10.1016/j.talanta.2024.126179.

[20]. Kaewjangwad, C., Somsiri, S., **Wangchuk, S.,** Saichanapan, J., Saisahas, K., Samoson, K., ... & Limbut, W. (2024). Smartphone-interfaced flow injection amperometric system for enzyme-free glucose detection using a palladium-PANI/carbon microsphere@carbon nanotubes modified electrode. *Electrochimica Acta*, *491*, 144292. (**Q1, IF 6.6**). https://doi.org/10.1016/j.electacta.2024.144292.

[21]. Promsuwan, K., Saichanapan, J., Soleh, A., Saisahas, K., Samoson, K., **Wangchuk, S.,** ... & Limbut, W. (2024). Nano-palladium-decorated bismuth sulfide microspheres on a disposable electrode integrated with smartphone-based electrochemical detection of nitrite in food samples. *Food Chemistry*, *447*, 138987. (**Q1, IF 8.8**). https://doi.org/10.1016/j.foodchem.2024.138987.

[22]. Promsuwan, K., Kareng, Y., Saichanapan, J., Soleh, A., Saisahas, K., Samoson, K., **Wangchuk, S.,** & Limbut, W. (2024). A novel 3D-printed portable electroplating device enhances latent fingerprints on metal substrates. *Talanta*, *272*, 125822. **(Q1, IF 6.1**). https://doi.org/10.1016/j.talanta.2024.125822.

[23]. Malarat, N., Soleh, A., Saisahas, K., Samoson, K., Promsuwan, K., Saichanapan, J., **Wangchuk, S.,** ... & Limbut, W. (2024). Electropolymerization of poly (phenol red) on laser-induced graphene electrode enhanced adsorption of zinc for electrochemical detection. *Talanta*, *272*, 125751. (**Q1, IF 6.1**). https://doi.org/10.1016/j.talanta.2024.125751.

[24]. Saraban, A., Promsuwan, K., Saichanapan, J., Soleh, A., Saisahas, K., Samoson, K., **Wangchuk, S.,** ... & Limbut, W. (2023). A Ternary Nanocomposite Based on Nano-Bimetallic Platinum/Nickel Decorated on Multi-Walled Carbon Nanotubes for Flow Injection Amperometric Detection of Promethazine. *Journal of The Electrochemical Society*, *170*(6), 067504. (**Q1, IF 3.9**). https://doi.org/10.1149/1945-7111/acdb9d.

[25]. Kongsuwan, A., Chuenjitt, S., Phua, C. H., **Wangchuk, S.,** Saichanapan, J., Saisahas, K., ... & Limbut, W. (2023). Sibutramine detection in weight-loss products using a sodium/phosphorus dual-doped carbon nanotubes modified electrode. *Microchemical Journal*, *190*, 108668. (**Q1, IF 5.1**). https://doi.org/10.1016/j.microc.2023.108668.

[26]. Promsuwan, K., Soleh, A., Samoson, K., Saisahas, K., **Wangchuk, S.,** Saichanapan, J., ... & Limbut, W. (2023). Novel biosensor platform for glucose monitoring via

smartphone based on battery-less NFC potentiostat. *Talanta*, *256*, 124266. (**Q1, IF 6.1**). https://doi.org/10.1016/j.talanta.2023.124266.

[27]. Promsuwan, K., Saichanapan, J., Soleh, A., Saisahas, K., Samoson, K., Phua, C. H., ... & Limbut, W. (2022). New electrode material integrates silver nanoprisms with phosphorus-doped carbon nanotubes for forensic detection of nitrite. *Electrochimica Acta*, *436*, 141439. (**Q1, IF 6.6**). https://doi.org/10.1016/j.electacta.2022.141439.

[28]. Chuenjitt, S., Kongsuwan, A., Phua, C. H., Saichanapan, J., Soleh, A., Saisahas, K., Samoson, K., **Wangchuk, S.,** ... & Limbut, W. (2022). A poly (neutral red)/porous graphene modified electrode for a voltammetric hydroquinone sensor. *Electrochimica Acta*, *434*, 141272. (**Q1, IF 6.6**). https://doi.org/10.1016/j.electacta.2022.141272.

[29]. Promsuwan, K., Saichanapan, J., Soleh, A., Saisahas, K., Phua, C. H., **Wangchuk, S.,** ... & Limbut, W. (2023). Polyaniline-coated glassy carbon microspheres decorated with nano-palladium as a new electrocatalyst for methanol oxidation. *Journal of Electroanalytical Chemistry*, *928*, 116995. (**Q1, IF 4.5**). https://doi.org/10.1016/j.jelechem.2022.116995.

[30]. Subba, J. R., & **Wangchuk, S**. (2022). Detection of adulterants in some common food items available in the Bhutanese market. *Bhutan Journal of Research and Development*, *11*(1). https://doi.org/10.17102/bjrd.rub.11.1.026.

SUBMITTED MANUSCRIPT

Peer-Reviewed Journal Articles

[1]. Saichanapan, J., Promsuwan, K., Saisahas, K., Soleh, A., Samoson, K., **Wangchuk, S.,** Abdullah, F; Kunalan, V; Abdullah, F; Abdullah, L.,F., A; Limbut, W, & Chang, K, H (2026). A Portable Voltammetric Sensor for Simultaneous Determination of Drug-facilitated Crime Substances (Xylazine and Ketamine) in Spiked Beverages. Journal of Forensic Sciences (Manuscript Number: **JOFS-26-545**)

PhD Thesis

[1]. **Wangchuk, S.,** Numnuam, A., Limbut, W. (2025). Development of Electrochemical Sensing Platforms Based on Functionalized Carbon Nanomaterials: Applications in Environmental Monitoring and Biomedical Analysis, Prince of Songkla University, Thailand.

RESEARCH REPORTS

[1]. **Wangchuk, S.** *Report: Water Source Sampling and Water Quality Testing for Dorjilung Hydropower Project Limited*. An exclusive technical report and research project funded by the Dorjilung Hydropower Project Limited (DHPP, 2026).

[2]. **Wangchuk, S.,** Tobgay, S., & Subba, J. R. *Final Report: Environmental baseline services for the 400 kV D/C East-West Transmission Line Project, Bhutan*. An exclusive technical report and research project funded by the BPC (January 2025).

[3]. **Wangchuk, S.,** Tobgay, S., & Subba, J. R. *Environmental baseline services for the 400 kV D/C East-West Transmission Line Project, Bhutan*. Technical post-monsoon season report: Research project funded by the BPC (November 2025).

[4]. **Wangchuk, S.,** Tobgay, S., & Subba, J. R. *Environmental baseline services for the 400 kV D/C East-West Transmission Line Project, Bhutan*. Technical wet season report, research project funded by the BPC (August 2025).

[5]. **Wangchuk, S.,** Tobgay, S., & Subba, J. R. *Environmental baseline services for the 132 kV D/C Gamri I & II Transmission Line Projects, Trashigang Dzongkhag, Bhutan*. Technical wet season report, research project funded by the BPC (July 2025).

[6]. Dendup T., Nidup, T., & **Wangchuk, S.** *Water Quality and Aquatic Ecology Assessment* for the 20 MW Yungichhu Hydropower Project. Technical report, research project funded by the DGPC (August 2021).

[7]. Nidup, T., Dendup, T., & **Wangchuk, S**. *Water Quality and Aquatic Ecology Assessment* for the 8 MW Thungdiri Hydropower Project. Technical report, research project funded by the DGPC (August 2021).

TRAINING/SEMINAR/WORKSHOP/CONFERENCE

Course/Workshop/Webinar Attended	**Institution/Agency, City and Country**	**Period From** (mm/yyyy) **To** (mm/yyyy)	
How to use Graphical Abstracts to Amplify your Research	Editage	22/07/26	22/07/26
The AI-Powered Document Health Check for Confident Submissions	Paperpal, Cactus Communications (Webinar)	10/06/26	10/06/26
Level up your publishing journey with ACS: Demystifying Peer Review – Part 2	ACS Publications (Webinar)	06/05/26	06/05/26
Level up your publishing journey with ACS: Demystifying Peer Review – Part 1	ACS Publications (Webinar)	29/04/26	29/04/26
Assessing the Impact of Institutional Accreditation on the Quality of Higher Education in Bhutan	Sherubtse College, Bhutan	22/04/26	22/04/26
Leadership Development Programme for RUB	RIGSS, Phuentsholing, Bhutan	01/03/26	07/03/26
How AI tools and human editorial expertise work together	Editage (Webinar)	17/02/26	17/02/26
Hook Readers Early: Writing Amazing Abstracts, Introductions, and Titles	Bentham Science (Webinar)	12/02/26	12/02/26
IMRaD Power! Writing Your Manuscript Section-by-Section	Bentham Science (Webinar)	15/01/26	15/01/26
Erasmus^{+} Regional Information Session	European Commission	04/12/25	04/12/25

for the Middle East and Asia	(Webinar)		
Why You Should Be a Peer Reviewer	Bentham Science (Webinar)	19/09/25	19/09/25
Strengthening Your Scholarly Communication-Resources & Best Practices	ACS Publications (Webinar)	14/08/25	14/08/25
Capacity Building Training on Air Pollution by Dr. Jas Raj Subba	Sherubtse College, Bhutan	30/08/25	30/08/25
Live Training Event: Research Writing for Maximum Readability and Reach	Bentham Science	21/08/25	21/08/25
Global Innovators: Breakthroughs in Sensors Research	ACS Publications (Webinar)	18/06/25	18/06/25
Spanning the Globe: Innovative Discoveries from Pioneering Researchers in Drug Delivery	ACS Publications (Webinar)	13/06/25	13/06/25
Advancing bRo5 Chemical Matter to the Clinic and Exploring Novel Kinase Inhibitors	ACS Publications (Webinar)	12/06/25	12/06/25
Global Innovators: Breakthroughs in Mass Spectrometry Research	ACS Publications (Webinar)	10/06/25	10/06/25
IEEE Authorship and Open Access Symposium: Tips and Best Practices to Get Published from IEEE Editors	IEEE Advancing Technology for Humanity	13/05/25	13/05/25
Green Hydrogen webinar: Exploring Bhutan's Green Hydrogen Potential	UNDP, Bhutan	04/04/25	04/04/25
Designing Impactful Online Learning Experiences	KnE Learn	7/02/25	7/02/25
Professional Development programme	IIT Kanpur, India	06/01/25	10/01/25
Envisioning a Hybrid Model of Peer Review: Integrating AI with Reviewers, Publishers & Authors	Editage Insights	24/09/24	24/09/24
Special Training Event: Why You Should Be a Peer Reviewer	Bentham Science (Online)	19/09/24	19/09/24
Ethics and Research Integrity	Dr De Gruyter (Organizer)	11/09/24	11/09/24
Live Event: "Effective Research Writing: The RIGHT way to WRITE"	Bentham Science (Online)	15/08/24	15/08/24
Increasing the Visibility of Scientific Research Through Graphic Design	Walailak University, Thailand	18/07/24	18/07/24
Live Event: "Go Figure! Secrets of Statistics and Data Presentation"	Bentham Science (Online)	12/06/24	12/06/24
Planning Your Publication: Pathways to Success	Bentham Science (Online)	16/05/24	16/05/24
Peer Review with Research Integrity in Mind	Dr Gareth Dyke (Online)	25/04/24	25/04/24
Peer review excellence: IOP training and certification	IOP Publishing	15/11/23	15/11/23
Publishing your research paper with Cambridge University Press	Cambridge University Press (Online)	08/03/25	08/03/25
Workshop on Breadth Modules	Sherubtse College, Bhutan	24/06/21	25/06/21
Wastewater and Emerging Pollutants: Unfolding Toxic Story	UNEP, Global Wastewater Initiative (Online)	09/06/21	09/06/21
Sustainable Nitrogen Management for	UNEP, Pakistan (Online)	04/06/21	04/06/21

Ecosystem Restoration			
Workshop on Education Reform	Sherubtse College, Bhutan	10/03/21	10/03/21
Scientific & Digital Learning	Sherubtse College, Bhutan	19/02/21	23/02/21
Capacity-Building PhD Program	AIT, Thailand	17/11/20	17/11/20
Qualitative research	Sherubtse College, Bhutan	17/02/20	18/2/20
ICT in Teaching and Learning	IIT Guwahati, India	20/01/20	23/01/20
Training on Gaussian	Sherubtse College, Bhutan	19/11/19	21/11/19
Introduction to University Learning & Teaching	Centre for University Learning & Teaching (CULT), GCBS	02/07/19	12/07/19

REFEREES

Prof. Dr. Warakorn Limbut
Deputy Director
Analytical Chemistry
Division of Health and Applied Sciences, Faculty of Science,
Prince of Songkla University, Hat Yai, Songkhla
E-mail: warakorn.l@psu.ac.th
Tel: +66 (0) 7428 8563
Fax: +66 (0) 7444 6681

Assoc. Prof. Dr. Chongdee Buranachai
Deputy Director
Analytical Chemistry
Division of Physical Science, Faculty of Science,
Prince of Songkla University, Hat Yai, Songkhla
E-mail: chongdee.t@psu.ac.th
Tel: +66 (0) 7428 8429
Fax: +66 (0) 7455 8841

Assoc. Prof. Dr. Apon Numnuam
Analytical Chemistry
Division of Physical Science, Faculty of Science,
Prince of Songkla University, Hat Yai, Songkhla
E-mail: apon.n@psu.ac.th
Tel: +66 (0) 7428 8747
Fax: +66 (0) 7455 8841

Dr. Jas Raj Subba
Head of Department
Analytical Chemistry
Department of Natural Sciences,
Sherubtse College, Royal University of Bhutan
E-mail: jasrsubba.sherubtse@rub.edu.bt
Tel: +975 17745478